Advance Labs I

	Experiment Theme				
	Fabrication and electrical characterization of MOS capacitors				
Experiment Start Date	15 December 2025	Report Submission Date		Submission Deadline	12 January 2026
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受領日	
受領者	

再提出

1					
2					
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5					
返却日		提出期限		提出場所	

受理

評点　　/10 点

原理・装置・測定手法は十分に理解されているか？				
a	b	c	d	e
実験データの解析は適切になされているか？				
a	b	c	d	e
実験結果は適切に提示されているか？				
a	b	c	d	e
考察の充実度は？				
a	b	c	d	e
目的、方法、結果、考察等の構成が適切か？				
a	b	c	d	e
丁寧に記述されているか？				
a	b	c	d	e

コメント

C. MOS Capacitor

Task 1: MOS Capacitor Fabrication Process

The ”Metal–Oxide–Semiconductor Field-Effect Transistor” (or MOSFET transistor) is a type of field-effect transistor (FET). In this experiment, we fabricated a MOS capacitor using a method involving the controlled oxidation of a p-type silicon wafer. The experiment was conducted in a ”clean-room” which is dust free and must only be entered with clean-suits (dust-repellent) and boots after taking an air shower..

MOS Capacitor Fabrication process that we followed during this experiment:
The main processes were Preparation, Oxidation, Aluminium deposition, Photo-resist coating, Patterning (Microfabrication) through Photolithography, Etching and Photo-resist peeling.

1. Preparation: A P-type Silicon wafer was taken and cleaned through the removal of meta, organic contamination, micro-particles etc.

Film formation:

2. Oxidation:. The Silicon Wafer was placed in a oxygen furnace ambient at 1000ºC using a quartz boat and clean quartz plate carefully. This results in a formation of a thin film of approximately 40 nm on top of the Silicon due to the following reaction: $Si + O2 \rightarrow SiO2$

3. Aluminum Deposition: The vacuum evaporation method was used to deposit an Aluminum (Al) thin film onto the Silicon wafer. A vacuum pump was used to create a vacuum in a glass bell jar before a voltage is applied and the Aluminum is heated. The wafer is placed inside the bell jar at the top facing downwards. Once the voltage is applied and the impurities in the Aluminum have evaporated settled onto a gate separating the Silicon wafer and the gate is opened and the pure Aluminum vapor is allowed to condense onto the wafer, forming a thin-film. The thickness of the aluminum film on the silicon wafer is 50 nm.

Microfabrication:

4. Photo-resist coating: On the second day, a light-reactive paint is applied on to the Silicon wafer with the SiO2 and Al films using a spin quarter.

5. Patterning (Microfabrication) through Photolithography: After placing the coated wafer on a heater to pre-bake, it is placed into a device where UV light is radiated onto the wafer through a photomask in the glass (with wiring patterns and the circuit drawn). The area except for the wiring patterns is now affected by the UV light, changing its properties from insoluble to soluble in the developing solution. The wafer is now

immersed in the developing solution for 1 minute and rinse for another minute to remove unaↄected photo-resist. It is then dried using the air-gun and heated again (post-bake). Since the unaffected photo-resist is resistant to etching solution for Aluminum, the wiring pattern is not affected by the etching solution.

6. Etching: The Silicon wafer is now dipped into a mixture of Phosphoric acid (H3P O4) and Nitric Acid (HNO3) that is heated using a hotplate for 1 minute and then rinsed in pure water for 1 minute. Th wafer is then dried using an airgun. It is observed that the area wihout the wiring now has the SiO2 layer exposed as the Al uncoated with photoresist gets dissolved in the acid through the following reaction: Al + 6HCl -> 2AlCl3 + 3H2

7. Resist peeling: The final step is to move to the ”yellow-filtered” -room, which uses yellow filter to cast out UV. We then follow a similar process as steps 4,5 is followed but without a photomask, exposing the remaining photoresist to UV light and dissolving it in the development solution. The wafer is then dipped into Acetone and Methanol for 5 minutes each to remove the leftover organic coating. Finally, it is rinsed in pure water for 5 minutes and dried with an airgun.

The fabrication process is explained in the figures below.

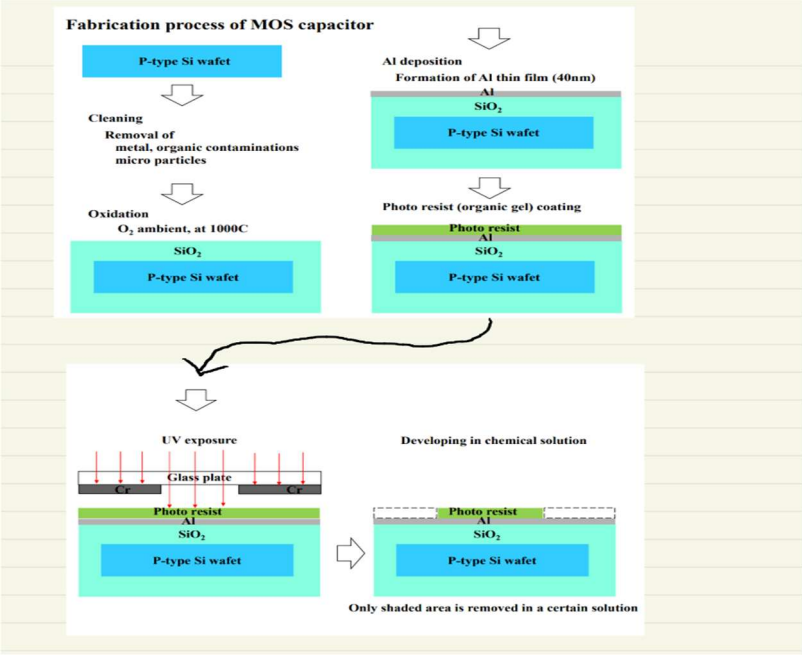


Figure 1 MOS Fabrication process

Task 2: Characteristic Analysis

Two methods were used to measure functioning and characteristics of the MOS capacitor.

The first is capacitor-voltage measurement, and the second one is Fowler-Nordheim Tunneling Current measurement method to measure current and breakdown voltage.

1. Capacitor-Voltage Measurement (C-V Measurement)

   The fabricated MOS capacitor was checked for functioning and characteristics by applying Voltage and Current. The following conditions were used:

   1) P-type silicon wafer is used
   2) Oxidation condition is 1000℃ for 30min.
   3) The electrode area used for CV measurement is 100μm x 100μm.
   4) Film thickness is calculated using the result of spectroscopic ellipsometry.
   5) The measurement frequency for CV measurement is 100KHz (high frequency).

Initially before we conducted the C-V measurement, the SiO_2 film thickness measured by ellipsometry is 25.54 nm. From the C-V measurement, we would also then compare the ellipsometry result with the SiO2 film thickness calculated from the results of the C-V measurement.

a) C-V measurement and graph

For MOS capacitor, C is dependent on V and the C versus V follows a nonlinear curve. The length of the area occupied by the Boron ions (due to p-type Si used) is L_D. The C-V characteristics of a MOS capacitor are calculated by taking the derivative of the Q-V graph where V is the DC voltage applied to the capacitor. A small AC charge is applied to change L_D and therefore, the charge Q.

1) If negative DC voltage is applied to the integrated circuit (to the gate), holes accumulate in the p-Si.

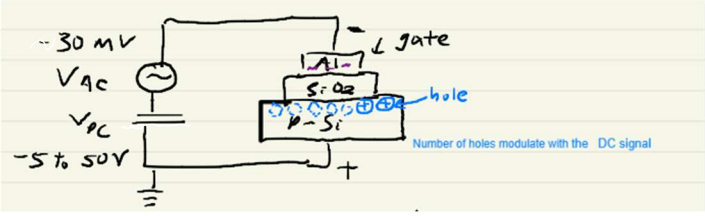


Figure 2

For a negative V_DC, the holes move upward, effectively creating a single capacitor arrangement. Therefore, the capacitance is flat for a negative voltage and decreases as the voltage becomes positive.

2) If positive DC voltage is applied to the circuit, effectively, resulting in an arrangement of two capacitors in series.

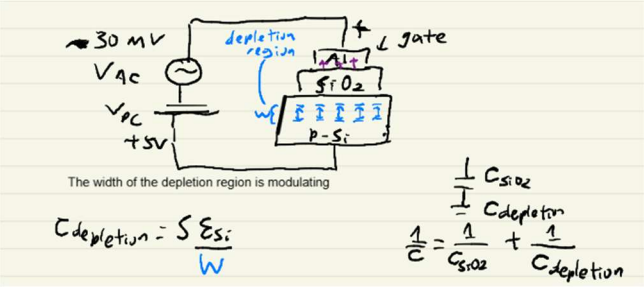


Figure 3

Applying a positive DC Voltage to the integrated circuit results in the neutral ions (holes) moving away from the source (downwards), resulting in the immobile Boron ions (negatively charged) to saturate in the upper area towards the SiO2 film. If we increase the DC voltage, the width of the depletion region becomes bigger.

b) Results and Discussion

The CV measurement characteristics of the MOS capacitor is recorded and the graph of the results is created using Excel.

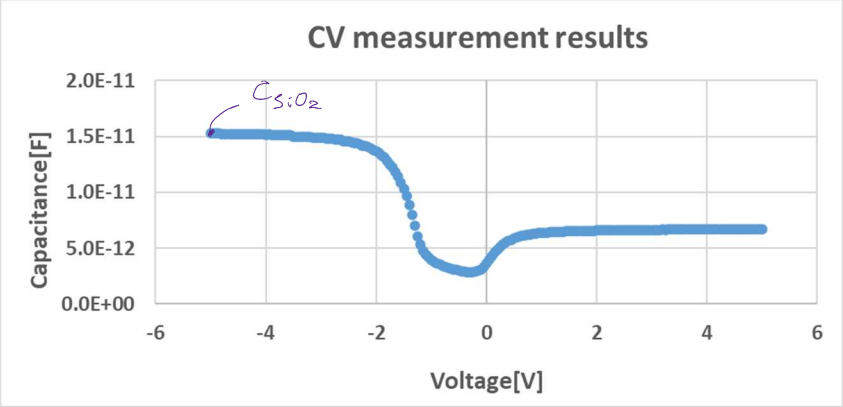


Figure 4 C-V Measurement results

After calculating the capacitance, the film thickness can be calculated by rearranging the formula:

$$C = \epsilon \frac{S}{d}$$

where C is the average capacitance of the flat part of the graph, ε is the dielectric constant for SiO2 (3.9 ε_0, S is the size of the electrode and d is the thickness of the SiO2 film.

The thickness is d= (ε S)/ C

C is calculated as the average capacitance, thus

$$C = \frac{[(1.5 \times 10^{-11}) + (6.6 \times 10^{-12})]}{2} = 1.1 \times 10^{-11}\ F$$

$$d = \frac{[(3.39 \times 8.85 \times 10^{-1})(100 \times 100 \times 10^{-12})}{1.1 \times 10^{-1}} = 30.0015\ nm = 30.0\ nm$$

The C-V curve is different from the expected C-V measurement curve graph for an ideal MOS capacitor due to defects. Al needs to be on both sides of the silicon wafer and there is also an oxide issue between SiO2 and Al, which is not so good.

If there are negative defects, the C-V graph will be shifted to the right. (Not passing though the origin). In contrast, if there are positive defects, the C-V graph will be shifted to the left. Since the C-V graph seems to be shifted to the left, there may be positive defects on the MOS transistor.

Task 3: Fowler-Nordheim tunneling current measurement

a) Principle

Electrons normally have an energy barrier to pass through from Al to Si through the SiO2 film. By applying a big voltage, the electrons can go through Al to Si by overcoming this energy barrier. This is known as Fowler-Nordheim tunneling.

i. If there is no voltage

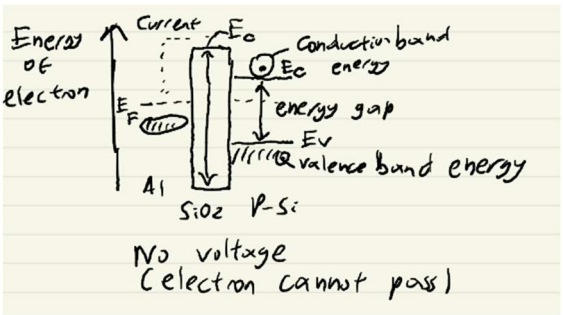


Figure 5

ii. If there is big voltage

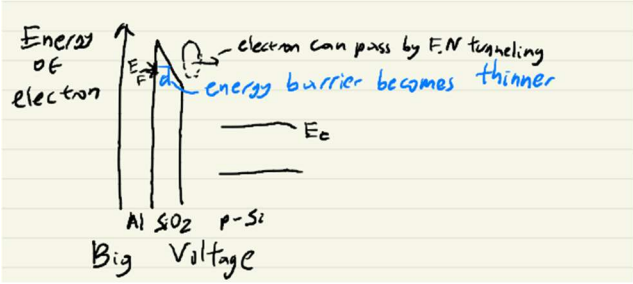


Figure 6

b) Results

Figure 7 shows the Current-Voltage (I-V) characteristics of the MOS capacitor. The current spikes for a greater negative voltage because of the F-N tunneling effect as more electrons can pass through. The current for less negative voltages (closer to zero) is largely constant but fluctuating due to noise.

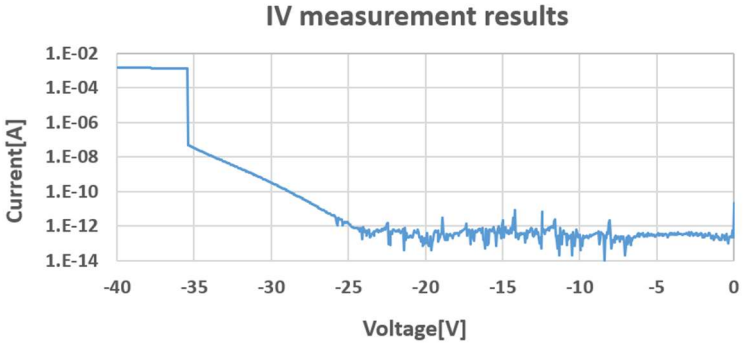


Figure 7 Current-Voltage measurement results

Task 4: Principles of Ellipsometry for measuring SiO2 film thickness

Spectroscopic ellipsometry is a non-destructive, noncontact, and non-invasive optical technique which is based on the change in the polarization state of light as it is reflected obliquely from a thin film sample. Ellipsometry uses a model-based approach to determine thin film, interface, and surface roughness thicknesses, as well as optical properties for thin films ranging in thickness from a few A to several tens of microns.
Also, spectroscopic ellipsometry can be performed either ex-situ or in-situ, in static or kinetic mode, for various application needs.

Spectroscopic ellipsometry measures ψ and Δ, both of which describe the output elliptical polarization state after linearly polarized light is reflected obliquely off of a thin film sample.
The parameters ψ and Δ are related to the complex Fresnel reflection coefficients according to: ρ = tan ψ eiΔ = rp / rs. After collecting ψ and Δ, a model representing the thin film structure must be built in order to determine thickness and/or optical constants.

An example of the setup and process for ellipsometry is in the figure below.

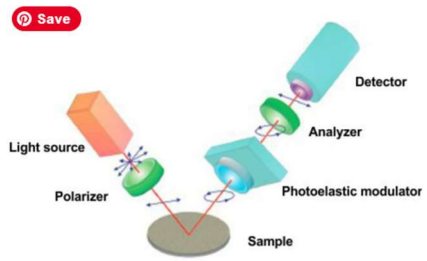


Fig.1: Optical setup of the UVISEL serie.

Figure 8

Typically, ellipsometers do not measure ψ and Δ directly. Instead, they measure functions of ψ and Δ. In the case of phase modulated ellipsometers, the three measureables are: Is, Ic, and Ic', which are functions of ψ and Δ according to Is = sin2ψ.sin Δ, Ic = sin 2ψ cos Δ, and Ic' = cos 2ψ. When combined, Is and Ic provide an accurate measurement of Δ over the full range from 0° to 360° and Is and Ic' provide an accurate measurement of ψ over the full range from 0° to 90°.

Spectroscopic ellipsometry is an indirect technique which does not measure thin film thickness and/or optical properties directly. In order to determine thin film thickness and/or optical properties, a model-based approach must be used.

Reference:
https://www.horiba.com/int/scientific/technologies/spectroscopic-ellipsometry/spectroscopic-ellipsometry/

Task 5: 実験の感想 (Impressions and Thoughts on the Experiment)
The MOS capacitor is a main part of MOSFET, which is used in integrated circuit (IC). The IC is used everywhere for electronics, including computers, phones, IC card (pasmo/

Suica), etc. Thus, this topic is integral to understand the phenomena of electronics.

The fabrication process was a very immersive and insightful experience, starting from bare silicon wafer. We had the opportunity to experience every fabrication step of MOS capacitor and MOSFET. Furthermore, the process was long, precise and complicated: requiring more apparatus/ equipment compared to the other themes we learned so far , and requiring cleanliness with precision and no contamination allowed. This is a rich novel, and fun experience. We had the opportunity to go to different lab rooms for different procedures. The professor is passionate in explaining the theory and the applications of semiconductor (n-type and p-type). The professor and the TA also thoroughly explained every process and experimental steps to us. From learning this lab theme of MOS capacitor, I developed a greater appreciation for the micro-electronic circuits used in devices and appliances all around us. This also increases my interest in studying semiconductors.